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Topic

Green Job Guarantee and Sectoral Inflation in Germany’s Heat Pump Market; 10-Year Forecast (2025-2035)

Introduction & Problem

Germany is committed to climate neutrality by 2045 to decarbonize residential heating (where 15% of the national emissions are generated) through rolling out heat pumps with the goal of 500,000 installations annually. To tackle the issue of job loss and job security in this green transition, a Green Job Guarantee (GJC) acts as a macroeconomic price stabilizer and ecological output driver to provide state-funded, ecologically focused jobs at a minimum fixed wage to those who are willing and able to work under this transition.

However, residential heat pump installation required highly certified technical labor (SHK technicians), which is in severe systemic shortage in Germany. As such, a GJG can trigger inflation in the cost of products in the sector in two ways. First, it takes the labor from the private sector when the exogenous wage floor is close to the private market-clearing wage for semi-skilled installation assistants, and then the private sector has to bid up wages to retain staff. Second, this wage-push cost spiral will affect consumers through markup pricing.

Research Question

Can a green job guarantee generate inflationary instability through sectoral labor mismatches during the residential green transition in Germany for the heat pump market?

Hypothesis

H0 (Null Hypothesis):
A Green Job Guarantee does not significantly increase inflationary pressure in Germany’s residential heat pump installation sector. Labor shortages and installation prices remain stable despite the implementation of a GJG.

H1 (Alternative Hypothesis):
A Green Job Guarantee significantly increases inflationary pressure in Germany’s residential heat pump installation sector through labor mismatches, wage competition, and higher installation costs.

Methodology

Baseline: The current German minimum wage holds; no public GJG program is implemented. The heat pump installation sector operates under persistent structural labor shortages and market-determined wage dynamics.

Scenario A: Low GJG Wage Floor and Low Skill Complementarity (the wage is set strictly equal to the national legal minimum wage). GJG projects focus strictly on low-skill environmental tasks.

Scenario B: High GJG Wage and Crowding Out of Technical Labor (the wage is set aggressively high to attract workers). The public sector initiates heavy infrastructure projects that directly compete for basic construction and civil engineering labor.

Scenario C: Accelerated GJG Demand and Scarcity-Induced Sectoral Stagflation (Government mandates heat pump roll-outs via regulatory shocks alongside a GJG, but the elasticity of substitution between GJC workers and certified SHK technicians is near zero.)

Table of Content

1. Introduction (1000 words)
    - Germany's 2045 climate goals
    - SHK technician shortage
    - GJG concept, RQ, Hypothesis
2. Methodology & Model (2500 words)
    - EIRIN model
    - equations (agent, parameter, variable)
3. Scenario Analysis (2500 words)
4. Discussion & Policy Implications (1500 words)
    - evaluate hypotheses
    - translate into policy
5. Conclusion (500 words)
    - Core findings
    - limitation, future study

A	B
(1) Introduction, (3) Scenario Analysis (5) Conclusion	(2) Methodology & Model, (4) Discussion

Feedback & Suggestions

In general though, the idea is ideal to show the crowding out effect in a more theoretical (top-down) model: in that I would argue that the time scale of the model is secondary. It's mostly showing the magnitude of differences between the different scenarios.

May I ask what kind of model you had in mind to work with?

As i have read paper based on EIRIN and DEFINE model, the differences as i understand is that;

EIRIN model is the model that looks into detail like sectors, finance, and stock-flow relations While **DEFINE model** is a broader and more macro model which is looking at the big picture they will not go into detail as EIRIN does cuz EIRIN they look at small elements and actors within.